



Vyomira Tech Educate Program

Advanced Cloud + DevOps + Kubernetes (15 Sessions | 3 Hours Each)



Session 1: DevOps Architecture & Cloud Foundations (Advanced)

Objective: Understand production-grade DevOps systems



Breakdown:

- 45 min: DevOps architecture deep dive
- 45 min: Cloud architecture patterns
- 90 min: Hands-on



Topics:

- DevOps maturity model
- CI/CD architecture (enterprise level)
- Monolith vs Microservices
- High-level architecture of AMI Factory (your system)
- Cloud models: IaaS, PaaS, SaaS (real use cases)



Hands-on:

- Setup AWS account + CLI
 - Design simple architecture diagram
 - IAM user creation with least privilege
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◆ Session 2: Advanced Linux for DevOps Engineers

Objective: Production-level Linux mastery

 Breakdown:

- 60 min theory
- 120 min hands-on

 **Topics:**

- Linux internals (process, memory, I/O)
- Systemd deep dive
- Advanced networking commands
- Package management (apt vs dnf vs yum)

 **Hands-on:**

- Create system services
 - Monitor processes
 - Debug CPU/memory issues
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◆ Session 3: AWS Deep Dive (Compute + Networking)

Objective: Build production-ready infra

 **Topics:**

- EC2 lifecycle & AMIs
- VPC deep dive (subnets, routing)
- Security groups vs NACLs
- Elastic IP, Load Balancer intro

Hands-on:

- Launch EC2 using your AMIs
 - Configure VPC manually
 - SSH + security group testing
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◆ **Session 4: Multi-Cloud Strategy (AWS vs GCP vs Azure)**

Objective: Industry-level multi-cloud understanding

Topics:

- GCP Compute Engine
- Azure VM + networking
- Identity differences (IAM vs Azure AD)
- Multi-cloud architecture patterns

Hands-on:

- Deploy VM in GCP
 - Deploy VM in Azure
 - Compare performance & configs
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◆ **Session 5: OS Hardening & Security Engineering**

Objective: Enterprise security practices

Topics:

- CIS benchmarks

- SELinux advanced modes
- Firewall architecture
- Vulnerability lifecycle

 Hands-on (your real tasks):

- Run base hardening script (dnf + SELinux)
 - Modify hardening script
 - Scan using Trivy / OpenSCAP
 - Fix vulnerabilities
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◆ **Session 6: Web Stack Engineering (LAMP & LEMP Advanced)**

Objective: Production web server setup

 Topics:

- Apache vs Nginx architecture
- Reverse proxy
- PHP-FPM tuning
- Virtual hosting

 Hands-on:

- Deploy LAMP stack
 - Deploy LEMP stack
 - Write automation scripts (your GitLab style)
 - Host test HTML
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◆ Session 7: Database Engineering & Hardening

Objective: Secure DB deployments

Topics:

- MariaDB hardening
- PostgreSQL config (pg_hba.conf)
- MongoDB setup
- Secure DB exposure

Hands-on:

- Automate mysql_secure_installation
 - Configure PostgreSQL remote access
 - Setup MongoDB admin user
 - Test DB connections
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◆ Session 8: Performance Tuning & Optimization

Objective: Production performance tuning

Topics:

- Nginx tuning (gzip, caching)
- System resource tuning
- Logs & monitoring basics

Hands-on:

- Optimize Nginx configs
- Load test website

- Analyze logs
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◆ Session 9: Application Deployment (Node.js + WordPress)

Objective: Real application hosting

Topics:

- Node.js deployment architecture
- PM2 clustering
- WordPress internals

Hands-on:

- Deploy Node.js app with PM2
 - Setup WordPress with Apache/Nginx
 - Configure permissions & DB
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◆ Session 10: SSL, Security & Automation

Objective: Production-grade HTTPS

Topics:

- TLS/SSL working
- Let's Encrypt automation
- Security best practices

Hands-on:

- Install Certbot
 - Automate SSL renewal
 - Secure WordPress site
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◆ Session 11: Docker (Advanced Containerization)

Objective: Production container usage

Topics:

- Docker architecture
- Image optimization
- Multi-stage builds
- Docker Compose

Hands-on:

- Install Docker securely
 - Build custom images
 - Containerize Node.js app
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◆ Session 12: CI/CD Pipeline Engineering

Objective: Enterprise pipeline creation

Topics:

- CI/CD design patterns
- GitHub Actions deep dive
- OIDC authentication (no secrets)

 Hands-on (your real flow):

- Create GitHub Actions pipeline
 - Integrate with AWS
 - Trigger builds on push
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◆ **Session 13: AMI Factory & DevOps Automation**

Objective: Real company-level automation

 Topics:

- AMI lifecycle
- Automation scripts
- Image versioning

 Hands-on (based on your work):

- Build AMI pipeline
 - Automate image creation
 - Implement “Golden Run”
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◆ **Session 14: Kubernetes (Practical Introduction)**

Objective: Container orchestration basics

 Topics:

- Kubernetes architecture
- Pods, Deployments, Services

- Scaling & self-healing

Hands-on:

- Setup Minikube
 - Deploy containerized app
 - Expose service
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Session 15: Final Project + Production Demo

Objective: End-to-end DevOps system

 Full 3 hours = project + demo

Project Flow:

- Code → GitHub
- CI/CD → Build
- Dockerize → Deploy
- Optional Kubernetes deploy

Hands-on:

- Full pipeline implementation
- Live deployment
- Troubleshooting

Demo:

- Present architecture
- Explain decisions